

CKS121/CSE 242

Software Requirements

Chapter 1: Introduction &
Fundamental of Requirements
Engineering (RE) Part 2

Learning outcomes

The students

- understand the influence of constraints on requirements.
- understand the interrelations between requirements engineering and organizational processes.
- learn the difference between problems and solutions in development processes.
- learn about the traditional systems analysis.
- understand the shortcomings of establishing requirements engineering as an early development phase.
- understand the advantages if requirements engineering is established as a continuous, project-spanning and cross-product activity.

Structure of this Lecture

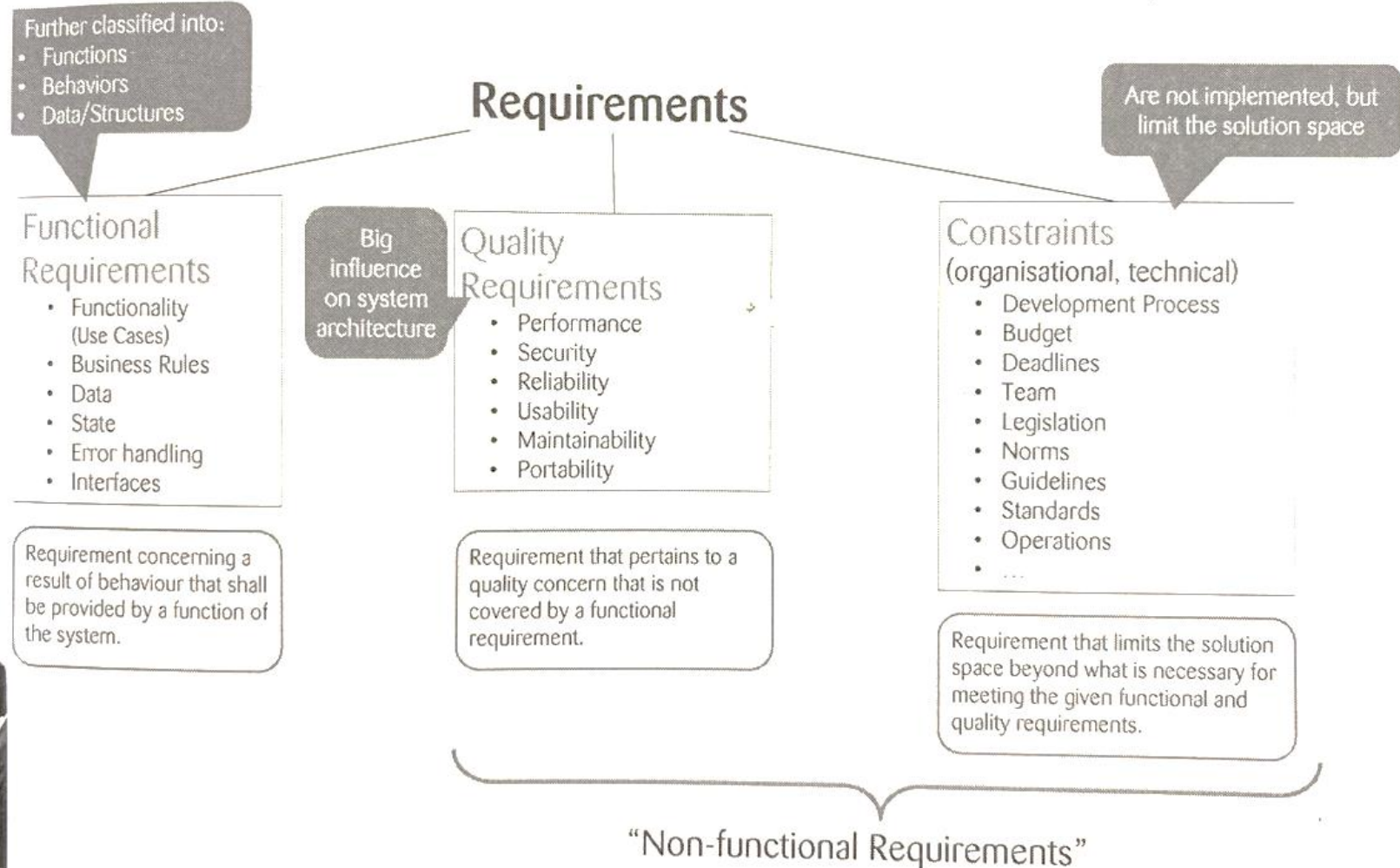
- **Influence of Constraints**
- Non-functional Requirements
- Requirements Engineering and Organizational Processes
- What vs. How in Development Processes
- Requirements Engineering as early Development Phase
- Continuous Requirements Engineering



Summary of Requirement Types

Revisit from
lecture 2

i More comprehensive quality models can be found in standards such as the Standard ISO/IEC25010:2011.



Influence of Constraints

Development Process



C-16: The effort for system development shall not exceed 480 person months.

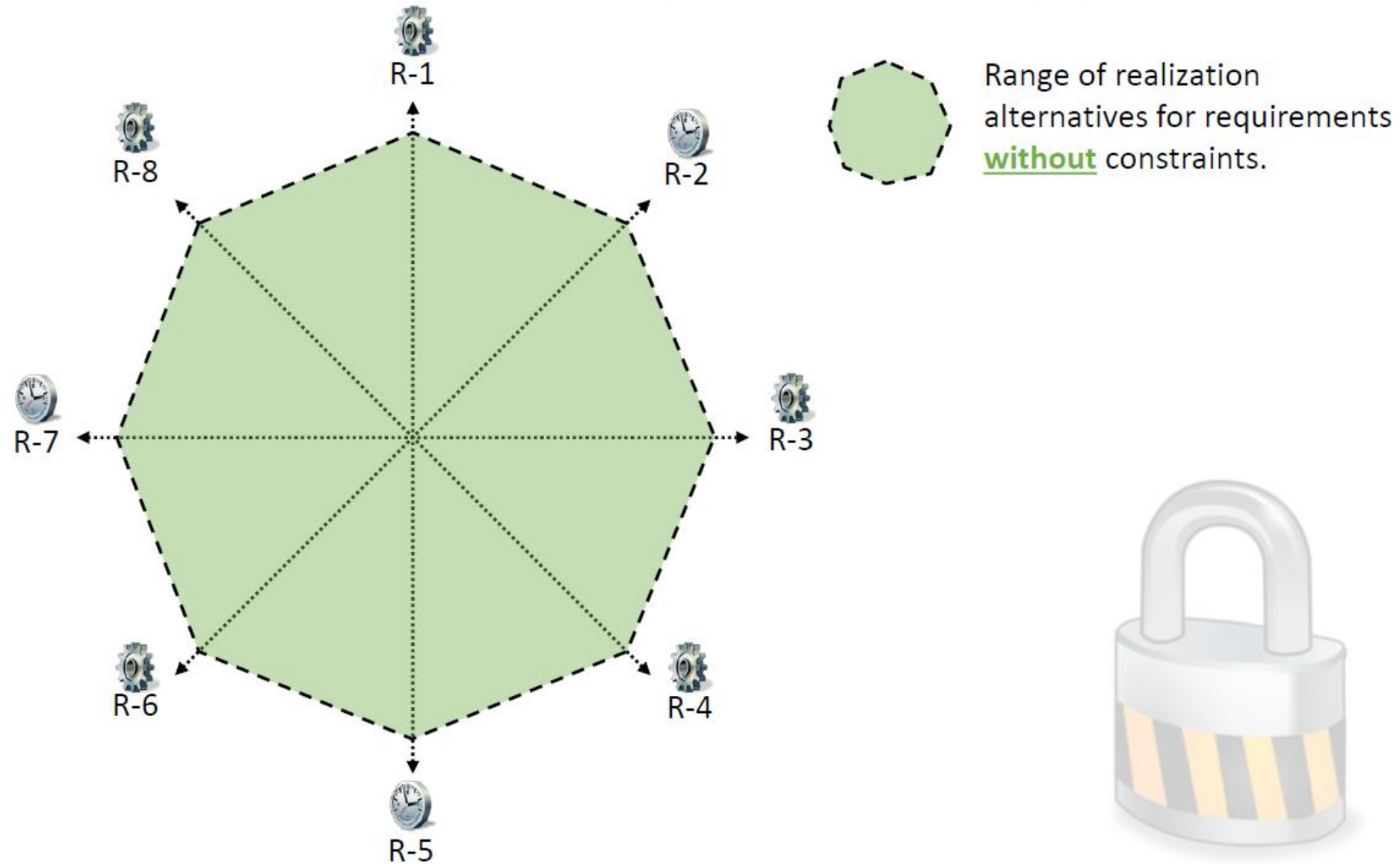
System



C-36: The electronic control unit in the vehicle interior shall work at temperatures from -10 to +50 degrees Celsius.

Constraints characterising the development process as well as constraints characterising the system usually affect the development process and the resulting product of the development process (i.e. the system itself)

Restrictions imposed by Constraints (1)



Restrictions imposed by Constraints (2)



R-3: The output shall be presented on a mobile phone.

Possible solutions, e.g.:

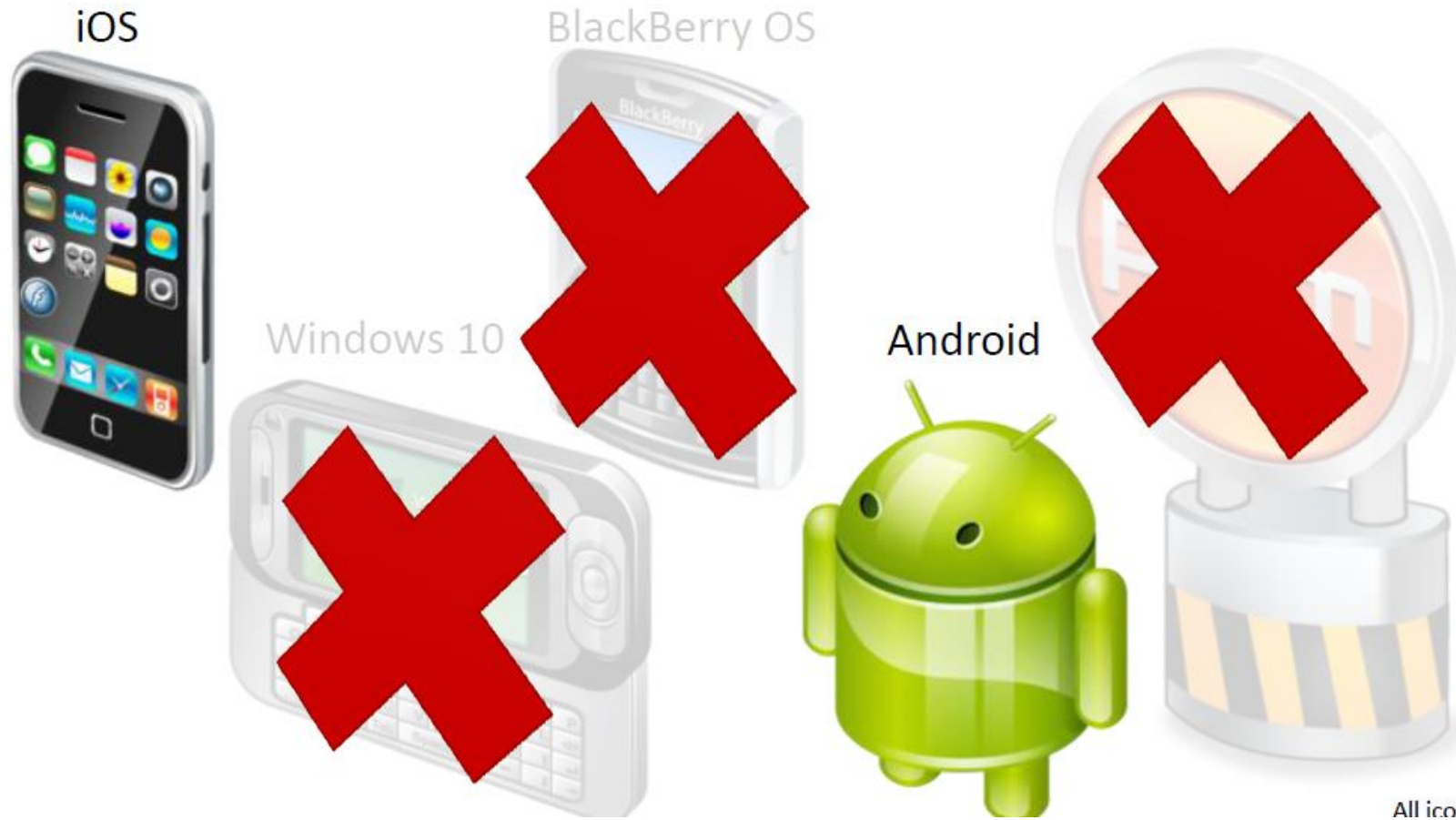


Restrictions imposed by Constraints (3)

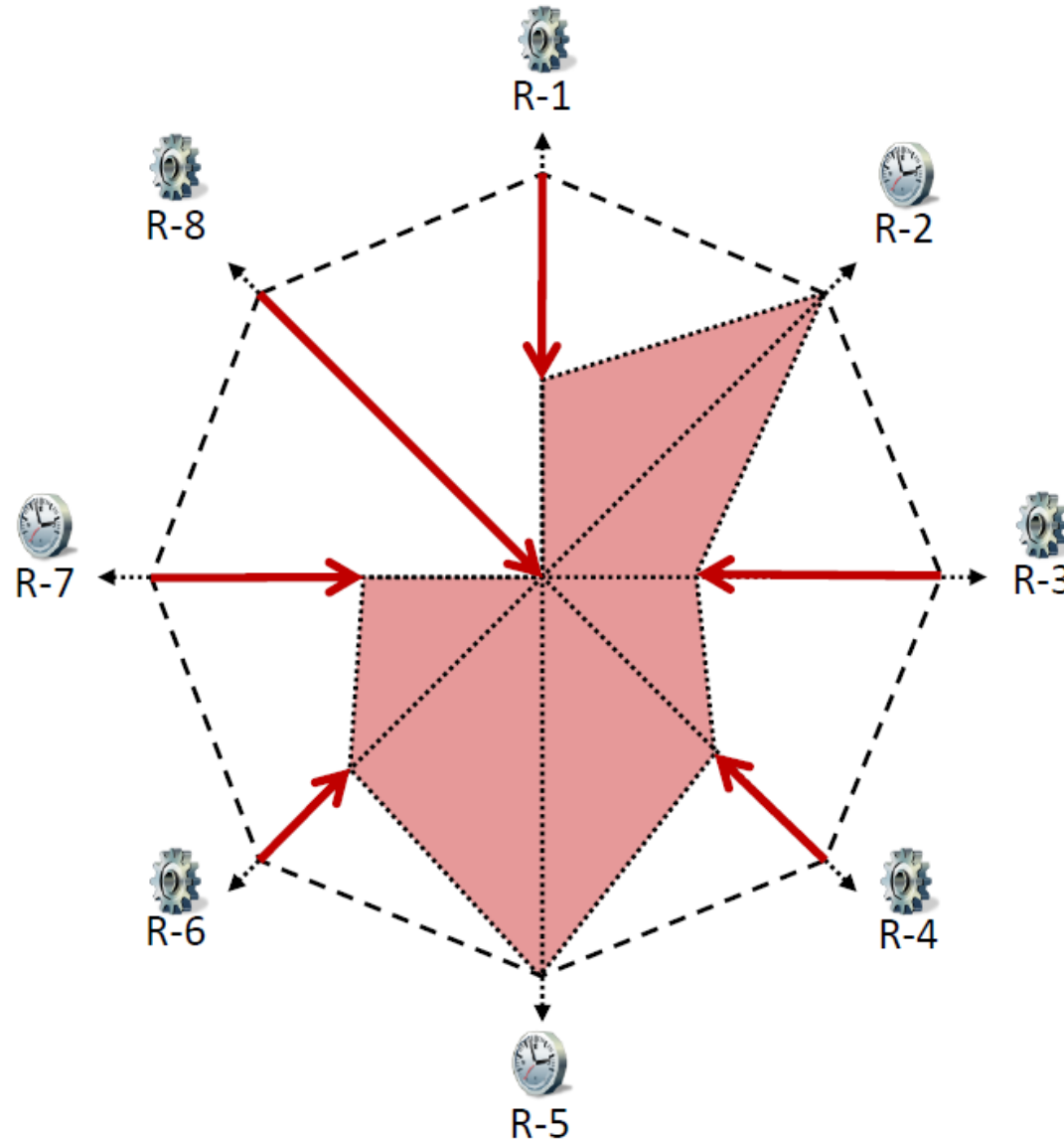


C-4: Only iOS and Android shall be supported.

Consequence: 2 out of 5 initial solutions left (40.0%)



Restriction on R-3 imposed by Constraint C-4



Range of realization alternatives for requirements without constraints.



Reduced range of realization alternatives for requirements caused by constraints.



Reduction of realization alternatives for a requirement caused by constraints.



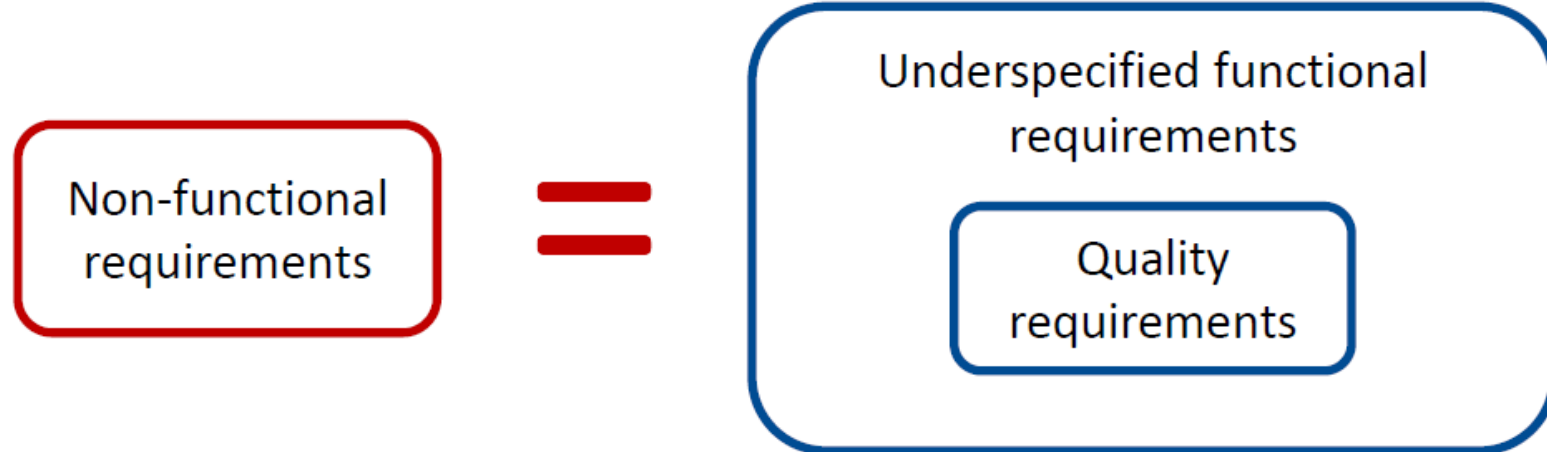
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Non-functional Requirements

- The term non-functional requirement is widely used in practice and literature.
- Whenever a requirement is called non-functional requirement, the requirement itself is typically insufficient understood.
- A so-called non-functional requirement is either
 - an underspecified functional requirement or
 - a quality requirement.



- There are only functional and quality requirements!
- Non-functional requirements do not exist!

Example for a “non-functional” Requirement



NFR-12: The system shall be secure.

→ NFR-12 is clearly an underspecified requirement.

More precise functional and quality requirements, which specify that “the system shall be secure” are:



R-12.1: Each user shall log in to the system with his user name and password prior to using the system.



R-12.2: The system shall remind the user every four weeks to change the password.



R-12.3: When the user changes his password, the system shall validate that the new password is at least eight characters long and contains alphanumeric characters.



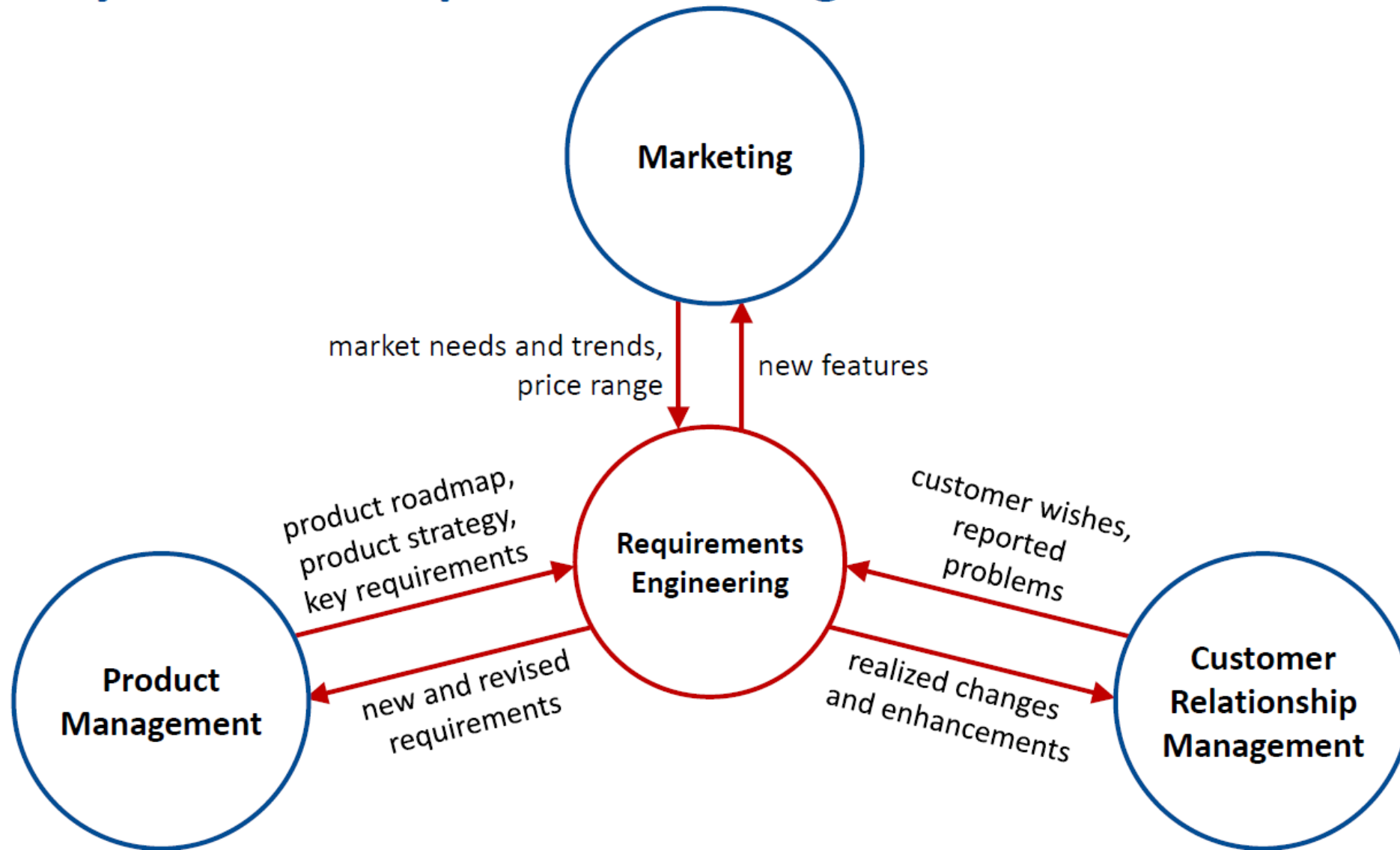
R-12.4: The user password stored in the system shall be protected against password theft.

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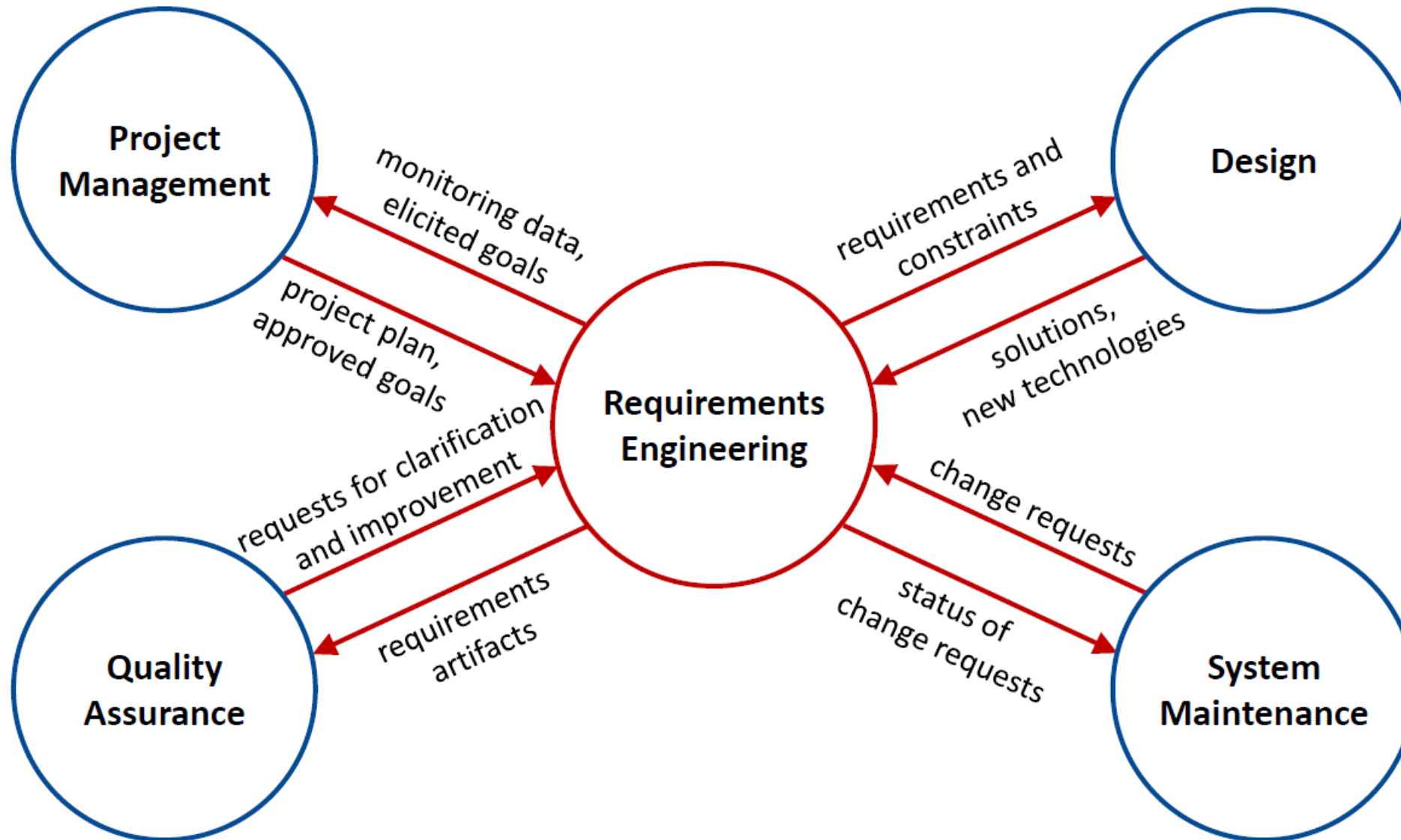
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Key Relationships to other Organizational Processes



Key Relationships to Development Activities



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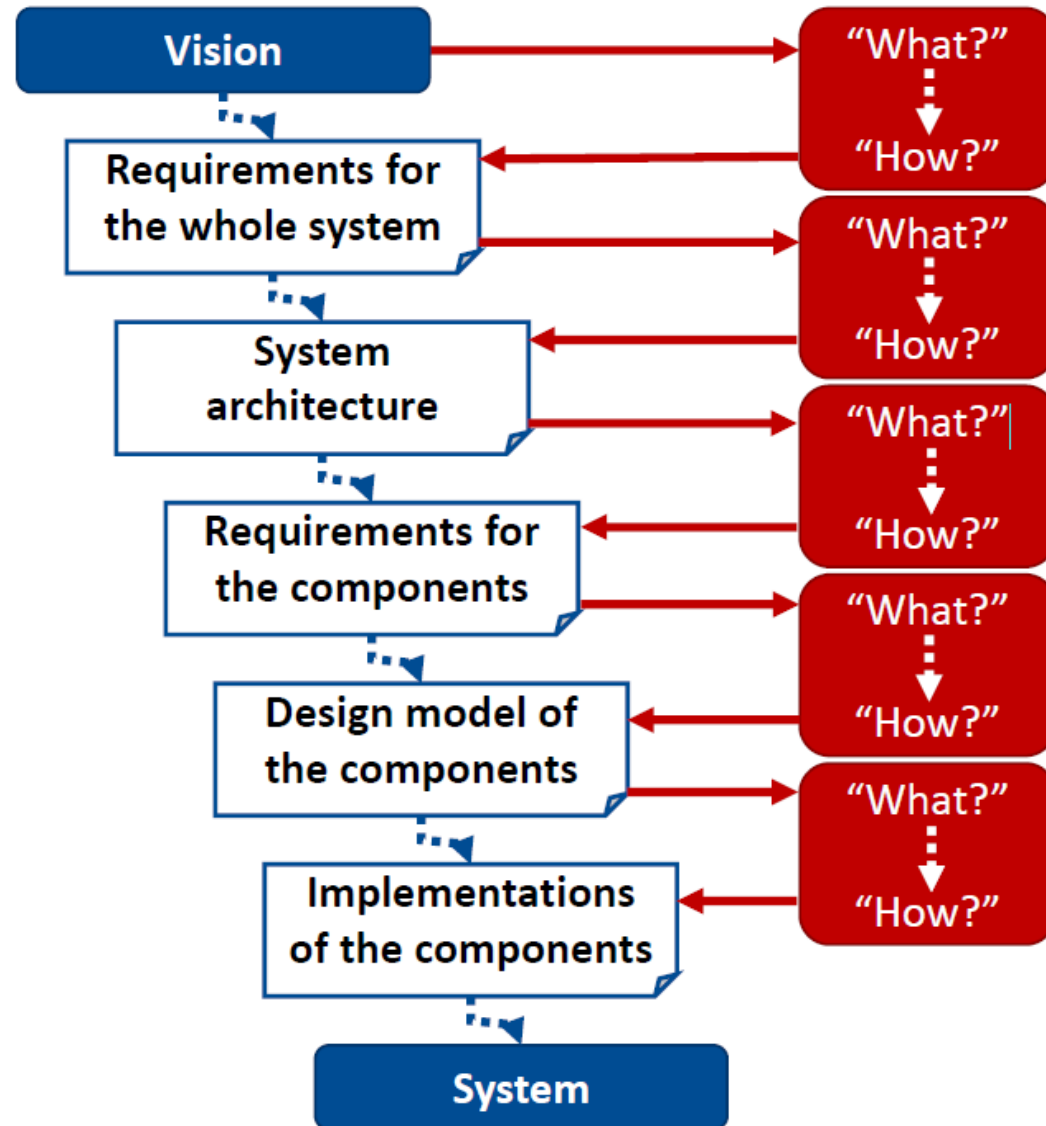
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Software Development Process: What vs How

- When we talk about **problems and solutions** in software development, we need to specify
 - “**what** system should be developed” → reflect “**software requirements**”
 - “**how** a system should be developed” → reflect “**software design**”
- Where software requirements answer the “**what**” question and software design answer the “**how**” question.
- In software development process, we can define several **what-how pairs** where each pair denotes a definition to the problem and corresponding solution description.

System Development Process



Within a Development Phase

Differentiation between “what” and “how” also occurs within a development phase.



Requirements Engineering Phase

What?	How?
• R-12: The navigation system shall allow the driver to enter the destination of the trip conveniently.	• R-12.1: When the driver starts a new trip, the navigation system shall display a roadmap of the area centered on the current position. • R-12.2: The navigation system shall allow the driver to scroll and zoom into the roadmap. • R-12.3: After the driver has selected a destination on the roadmap, the system shall allow the driver to edit the destination details (city, street and house number).

Refinement of Requirements

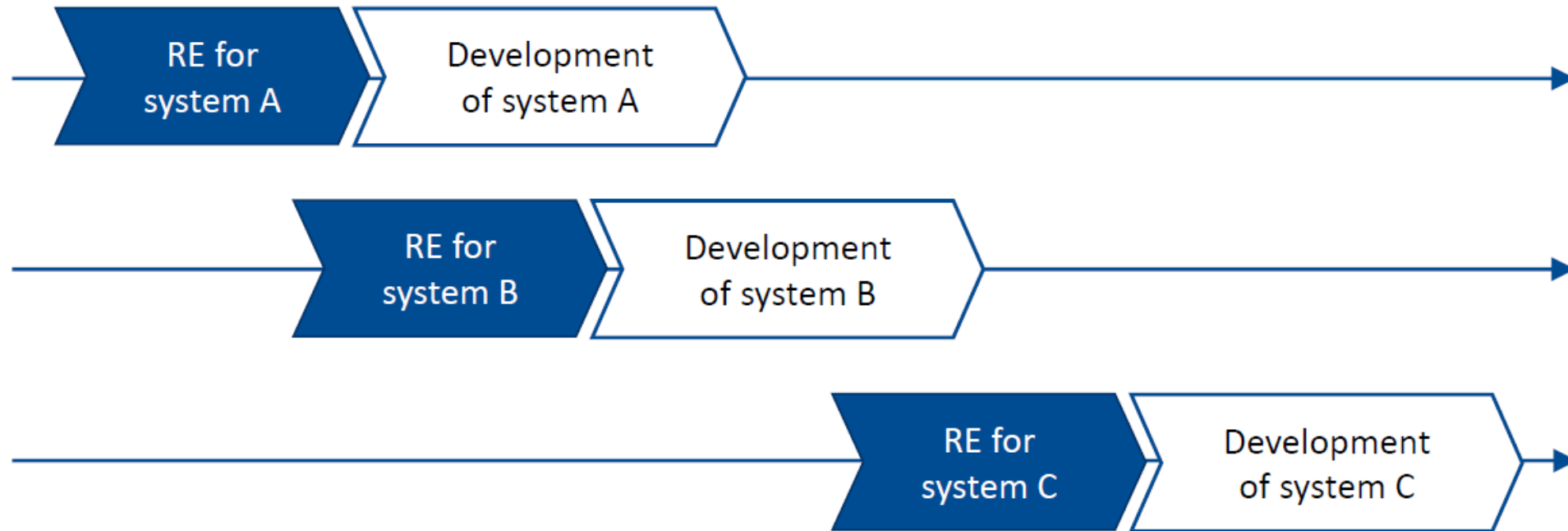
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Traditional Lifecycle and Process Models

Requirements engineering (RE) is regarded as the early (first) phase of system development.

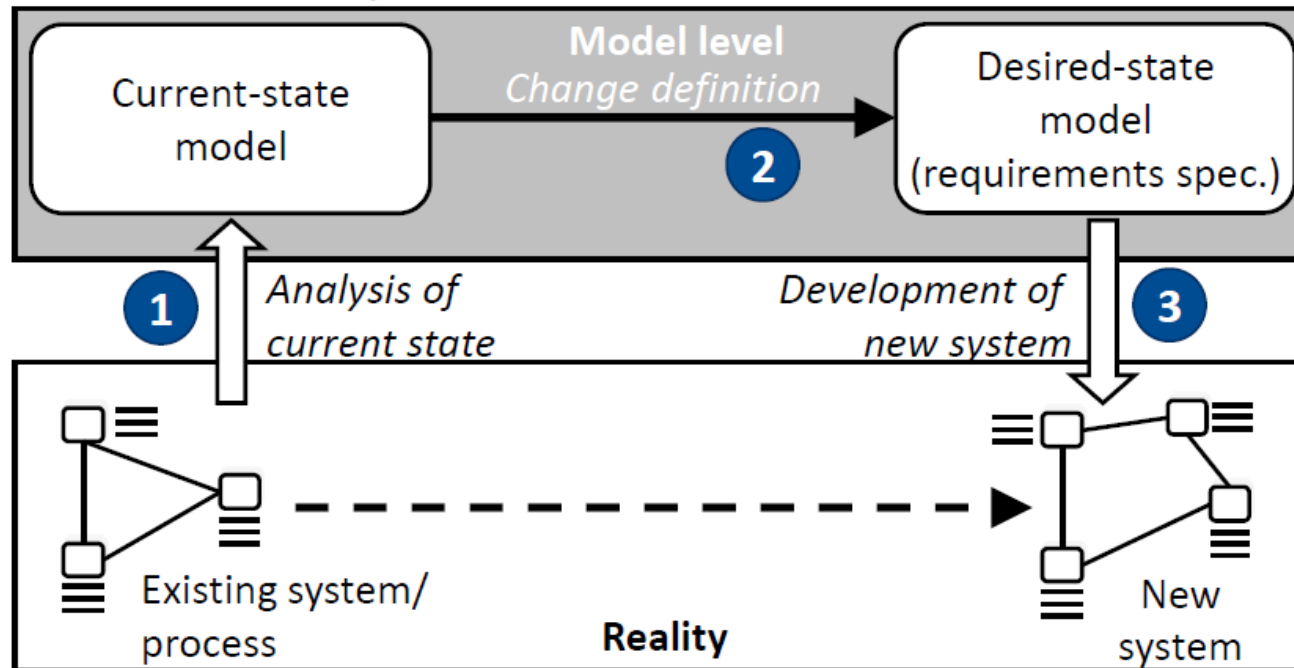


80's – early 90's: Traditional System Analysis

- Systems analysis aims at understanding and defining requirements of existing systems or processes in terms of function, data and behaviour.
- Typically the new system (partly) automates and thereby (partly) replaces existing systems and processes.
- Well known system analysis approaches are
 - Structured Analysis by DeMarco. [DeMarco 1978]
 - Essential System Analysis by McMenamin and Palmer. [McMenamin and Palmer 1984]

Typical System Analysis Activities

1. By analysing existing systems and processes a requirements model of the current (existing) system is created (current state model).
2. Based on the current state model intended changes are defined resulting in the desired state model. This model specifies the requirements for the system to be developed.
3. Realization of the new system based on the desired state model.



Shortcomings (1)

Requirements engineering as an early development phase has several shortcomings:

- **No continuity**
 - Requirements engineering is only performed for a particular time (as early phase in development).
 - Changes that occur later, e.g. during development, are not reflected in the requirements.
→ requirements become out-of-date.
- **Analysis of the current state**
 - Time consuming current state analysis at the beginning of each project.

Shortcomings (2)

- **No systematic requirements reuse**
 - Reuse of requirements across project and product boundaries is not systematically supported.
 - Requirements have to be developed from scratch, even if a large set of requirements are relevant for more than one product/project.
 - Reuse happens, if at all, in an ad hoc manner.
- **Narrow focus**
 - Focus of requirements engineering is typically restricted to the system under development.
 - Important opportunities for product innovation are lost.

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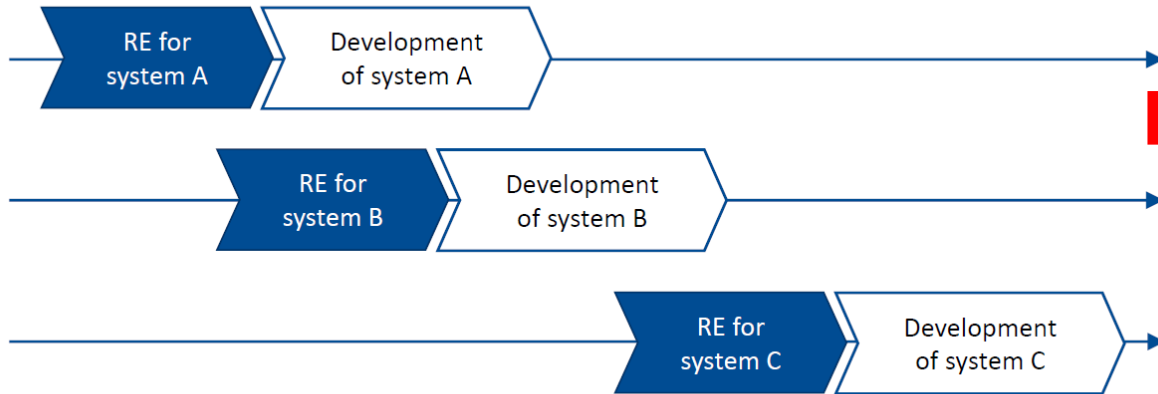
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Requirements Engineering as early Development Phase

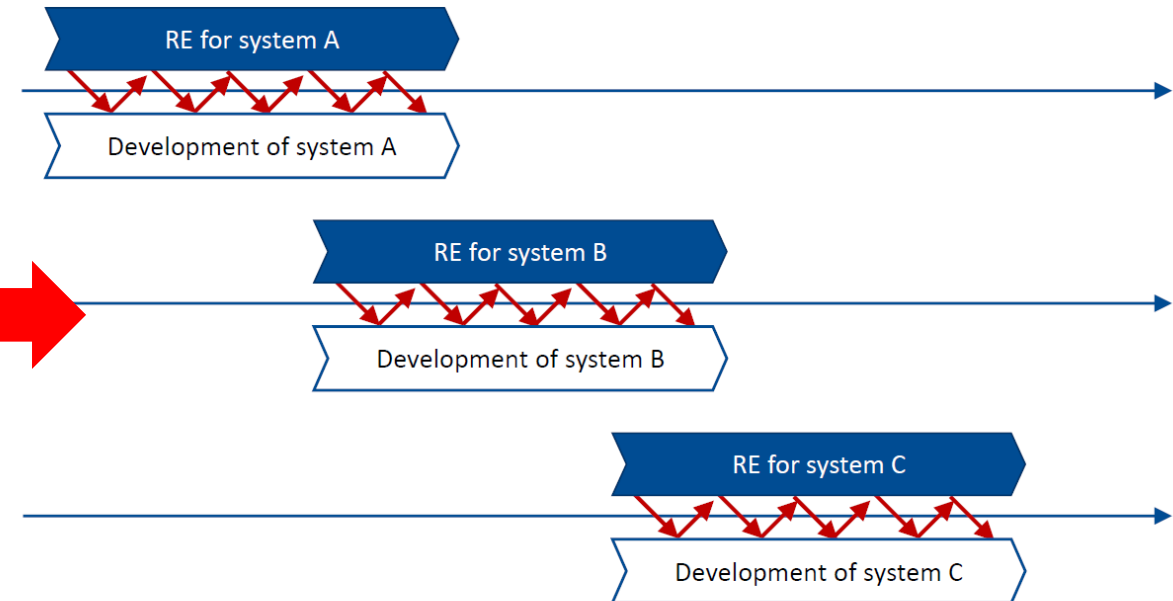
Traditional Lifecycle and Process Models

Requirements engineering (RE) is regarded as the early (first) phase of system development.



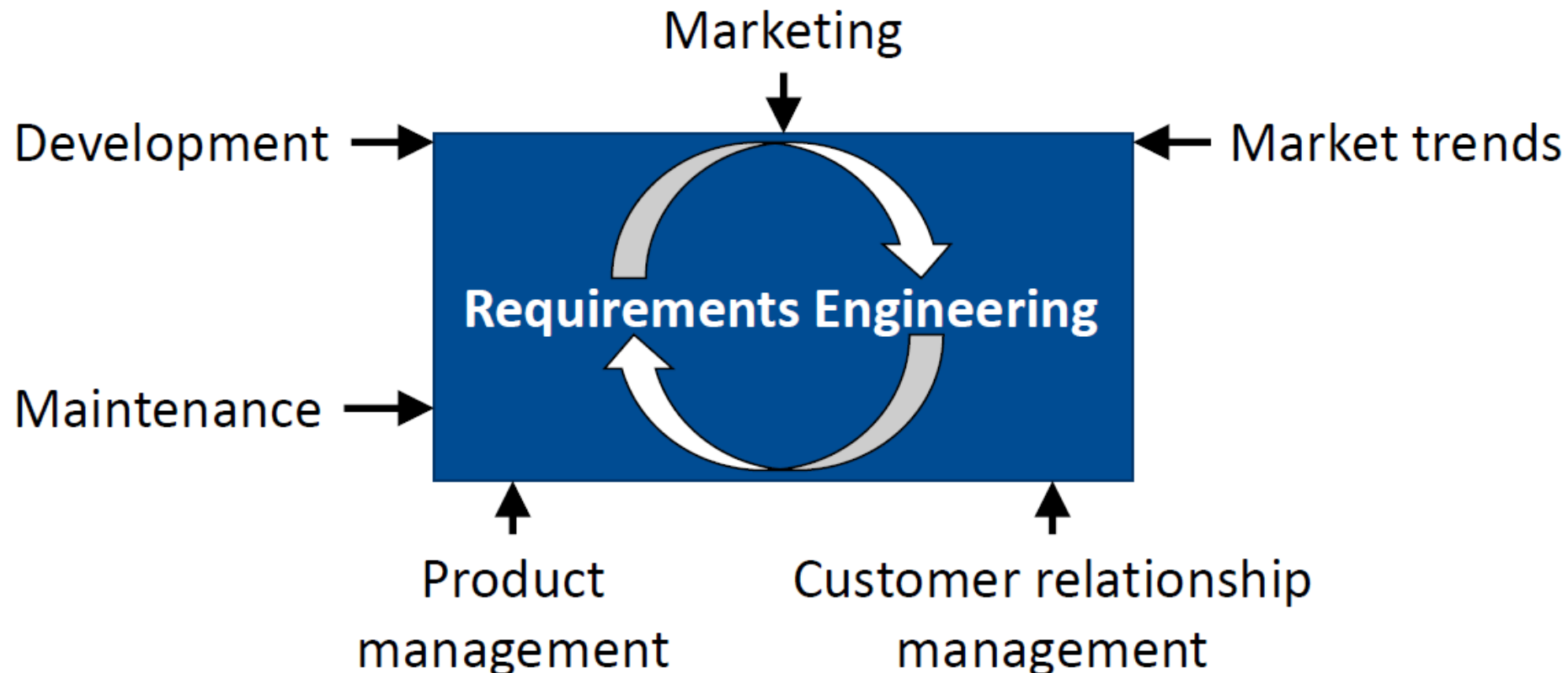
Continuous Requirements Engineering

RE as Cross-Lifecycle Activity



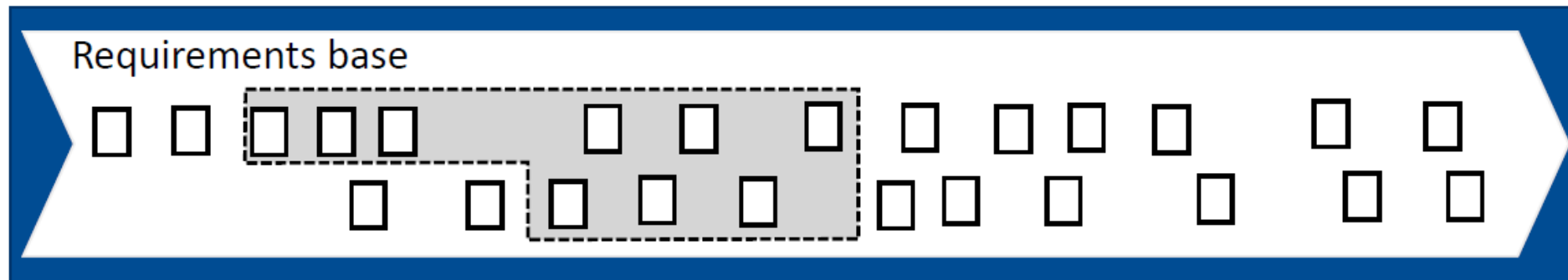
RE as Cross-Project and Cross-Product Activity (1)

- RE is implemented within an organization as continuous activity across project and product boundaries.
- The inputs from various sources are, e.g.:

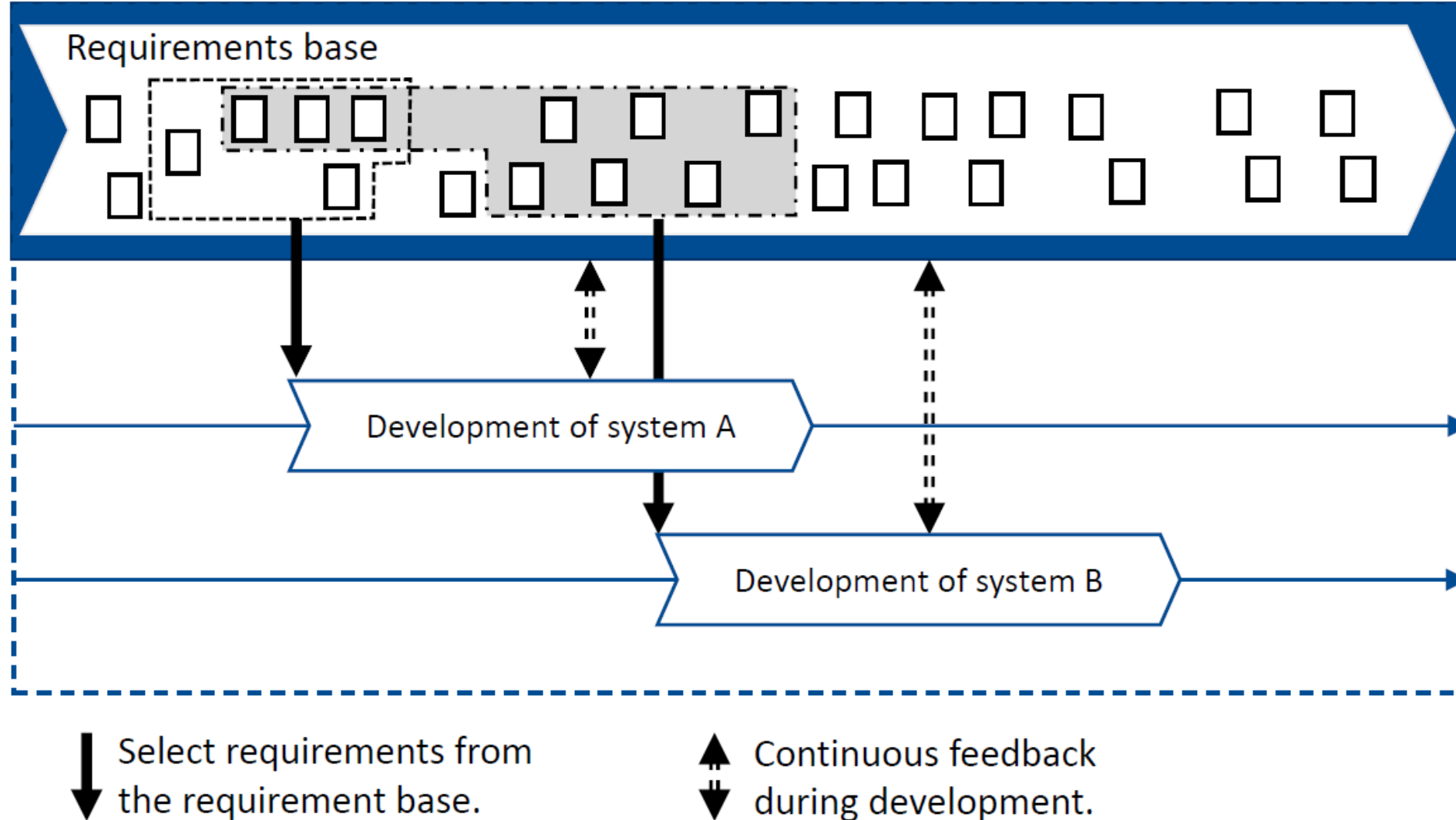


RE as Cross-Project and Cross-Product Activity (2)

- Continuously creates, changes and deletes requirements.
- The requirements base contains requirements under development as well as agreed, complete and correctly specified requirements.
- At a given point in time a set of requirements to be realized in the next system or system release is selected from the requirements base.



RE as Cross-Project and Cross-Product Activity (3)



Advantages (1)

- **Systematic learning process**
 - Institutionalized learning process in which the stakeholders involved continuously extend and improve their understanding of the requirements.
 - Sharing of requirements across projects.
- **Requirements are always up to date**
 - Up-to-date requirements base across products and projects.
 - Ensures that changes are integrated into the requirements base.
- **Shorter product development times**

Time-consuming analysis of the current state at the beginning of each project is prevented.

Advantages (2)

- **Reuse across projects and products**
Facilitates the reuse of requirements and related development artefacts across projects and products.
- **Clear responsibilities**
One or multiple people are explicitly responsible for the development and the management of the requirements across projects and products.

Summary

- Non-functional requirements are underspecified functional requirements and/or quality requirements!
- Requirements engineering is embedded within organizational processes.
- Traditional system analysis is typically performed as an early phase of system development.
- It defines the desired requirements of a new system in terms of functions, data and behaviour based on an analysis of existing systems or processes.
- Requirements engineering as an early phase of system development has significant shortcomings.
- Continuous requirements engineering as cross-project and cross-system activity overcomes these shortcomings.

Questions for Self-Assessment

- What is a non-functional requirement?
- Which types of constraints do you know?
- Name the key interactions between requirements engineering and organizational processes.
- What are key interactions between requirements engineering and other development activities?
- What is the main goal of “system analysis”?
- What are the shortcomings of requirements engineering as an early development phase?
- Describe the concept of requirements engineering as a cross-lifecycle activity. What are the advantages of continuous requirements engineering?

Questions ??

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